MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 11745243
POSITION-4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Position-4.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11745243	Circuit Card Assembly, Position-4

(Copies of specification, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition. This assembly is the fourth of four (4) single ended printed circuit boards consisting of operational amplifiers, diodes, and gates necessary to provide proper target position information.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11745243.

3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

Source: https://assist.dla.mil - Downloaded: 2026-10-22 19:02:52
Check the source to verify that this is the current version before use.

3.2.1.1 Power Supply.- The requirements of 3.2 shall be met with the following power supply voltages:

a. +5 VDC

b. +10 VDC

c. +15 VDC

d. -15 VDC

e. (0 to +5) VDC

f. (0 to -5) VDC

3.2.1.2 Nand Gate.- Input of pins 24, 23, and 20 provide the inputs to the triple input nand gate; the gate output is yielded on Pin 36 as shown in Table 1.

3.2.1.3 Operational Amplifier/Nor Gate Network #1. When both inputs Pins 25 and 23 are within the voltage range between -1.15V and +1.15V the output Pin 16 shall be Logic 1, otherwise output Pin 16 shall be Logic 0.

3.2.1.4 Operational Amplifier Network #1.- Inputs are provided at Pins 41, 40, and 39 and the outputs are yielded on Pins 28 and 38 as shown in Table 2.

3.2.1.5 Operational Amplifier Network #2.- Inputs are provided at Pins 34, 37, and 35 and outputs are yielded at Pins 30 and 3 as shown in Table 3.

3.2.1.6 Operational Amplifier/Nor Gate Network #2.- Inputs are provided at Pins 41, 40, 39, 34, 37, and 35 and the outputs are yielded at Pins 10 and 8.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11745243 and the engineering parts list.

3,3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11745243 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS.

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specififed in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and counter testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment. (See Figure 1)

4.2.1.1 Power Supply.- The power supplies shall be as follows:

a. +10,000 VDC ±1m VDC, 100mA or greater and a ripple of less than 10mVRMS or 100mV peak to peak.

b. +15,000VDC ±1mVDC, 100mA or greater and a ripple of less than 10mVRMS or 100mV.

c. +20.0VDC ± 0.1. VDC, 100mA or greater and a ripple of less than 10mA or greater and a ripple of less than 10mVRMS.

d. -20.0VDC ± 0.1 VDC, 100mA or greater and a ripple of less than 10mVRMS.

e. (0.000 to 7.000)VDC ± 1mVDC, 100mA or greater and a ripple of less than 10mVRMS.

f. (0.000 to -7.000)VDC ± 1mVDC, 100mA or greater and a ripple of less than 10mVRMS.

4.2.1.2 Oscilloscope.- Rise time of scope plus associated measuring probe of 50 nano seconds or less, sweep period of 50 nano seconds per centimeter or greater, input resistance greater than 8 megohms, nano input capacitance less than 25 picafarads.

4.2.1.3 Dummy Loads.- The dummy loads shall be triple (3)-input nand gates, one (1) type SN54L10 and three (3) type SN5410 or their equivalents.

4.2.1.4 Digital Voltmeter.- The digital voltmeter shall have a voltage accurancy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

a. Temperature between 60° F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power Supply.- All requirements shall be met with the following power supplies:

a. +5.000 VDC ± 1mVDC

b. +10.000 VDC + 1mVDC

c. +15.0VDC ± 0.1VDC

d. -15.0VDC ± 0.1VDC

e. (0.000 to 7.000)VDC ± 1mVDC

f. (0.000 to 7.000)VDC ± 1mVDC

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and _5.00 VDC.

4.2.2.3 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3.1 Nand Gate Test.- The nand gate shall function as shown in Table 1.

TABLE 1

Input (Logic)			Output (Logic)
Pin 24	Pin 23	Pin 20	Pin 36
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

4.2.2.3.2 Operational Amplifier/Nor Gate Network #1 Test.- When input Pins 25 and 33 are activated as described, output Pin 16 shall function as stated in the below test procedure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

Operational Amplifier/Nor Gate Network #1 Test Procedure.

NOTE: All voltages are D.C.

With 0V ± 1mV applied to Pins 25 and 33, output Pin 16 shall be Logic 1. Incrementally increase Pin 25 voltage, when it exceeds 1.150V ± 10mV, ouput Pin 16 shall switch from Logic 1 to Logic 0. Pin 16 shall remain at Logic 0 until Pin 25 falls below 1.150V ± 10mV, then switch back to Logic 1.

Return Pn 25 to 0V ±1mV. Incrementally decrease Pin 25 voltage, when the voltage falls below -1.150V ± 10mV, output Pin 16 shall switch from Logic 1 to Logic 0. Pin 16 shall remain at Logic 0 until Pin 25 voltage increases above -1.150V ± 10mV, then switch back to Logic 1.

Return Pin 25 voltage to 0V ± 1mV. Repeat the above procedure but vary Pin 33 voltage instead of Pin 25, output Pin 16 shall function in the same manner as described above.

4.2.2.3.3 Operational Amplifier Network #1 Test.- When inputs are applied to Pins 41, 40, and 39, output Pins 38 and 28 shall function as shown in Table 2.

TABLE 2

Inputs (VDC)			Output (VDC)	
Pin 41	Pin 40	Pin 39	Pin 38	Pin 28
1V ± 1mV	0V ± 1mV	0V ± 1mV	-4.00V ± .1B	5.48V ± .2V
0V	-IV	0V	5.534V ± .IV	-7.58V ± .2V
0V	0V	5V	-4.01V ± .IV	5.49V ± .2V
10V	0V	0V	-10.0V ± /5V	Not applicable

4.2.2.3.4 Operational Amplifier Network #2 Test.- When inputs are applied to Pins 34, 37 and 35, output Pins 30 and 3 shall be as shown in Table 3.

TABLE 3

Inputs (VDC)			Outputs (VDC)	
Pin 34	Pin 37	Pin 35	Pin 30	Pin 3
IV ± 1mV	5V ± 1mV	0V ± 1mV	-3.00V ± .1V	4.11V ± .2V
0V	5V		0V ± .1V	0V ± .2V
0V	5V		8.00V ± .1V	-10.96V ± .2V
-5V	5V		+10V ± .5V	Not applicable

4.2.2.3.5 Operational Amplifier/Nor Gate Network #2 Test.- When the inputs are applied as described below to Pins 41, 40, 39, 34, 37, and 35, output Pins 10 and 8 shall function as shown in the below test procedure.

Operational Amplifier/Nor Gate Network #2 Test Procedure

NOTE: All voltages are D.C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

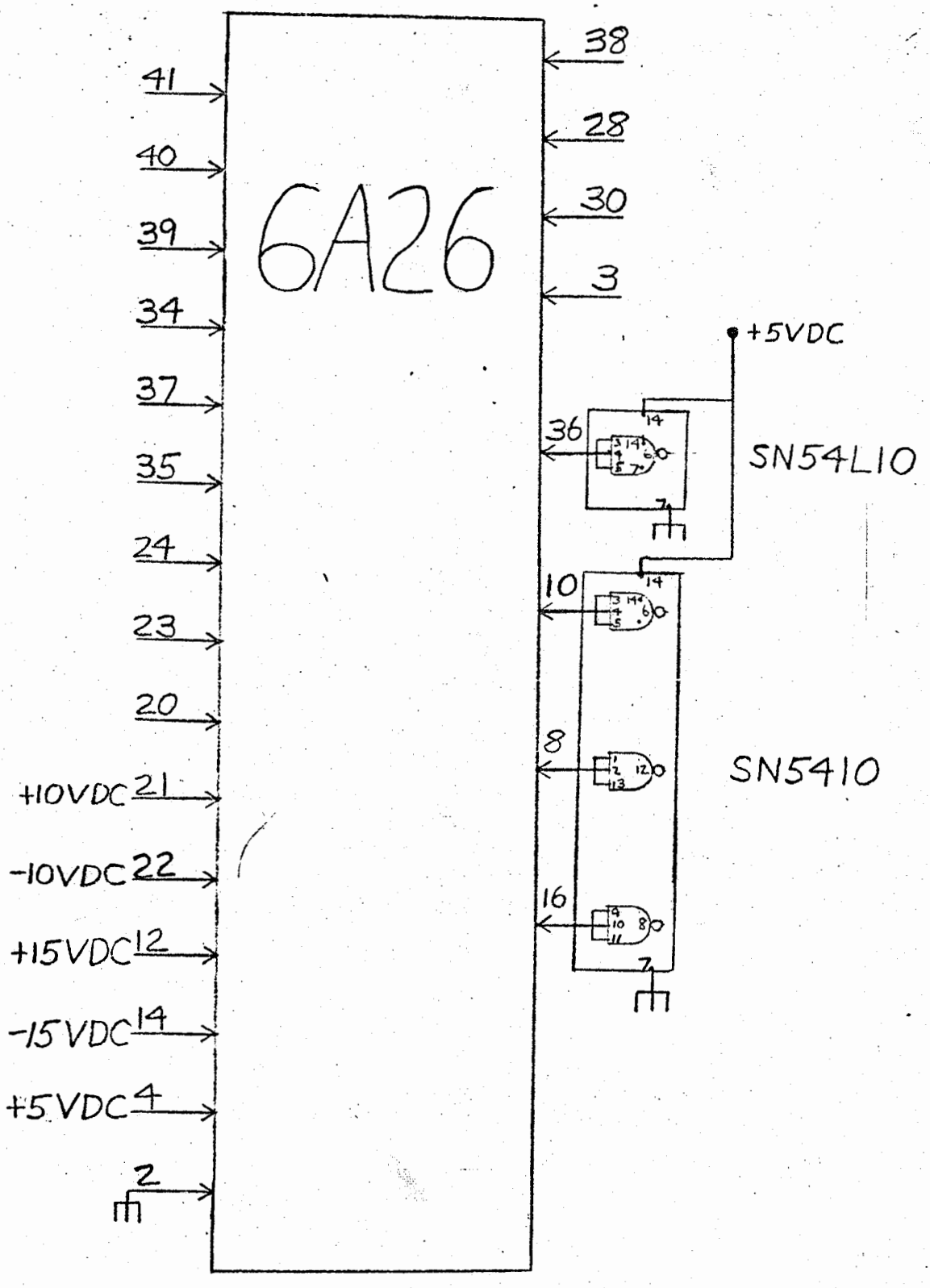


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

Apply 5V ± 1mV to Pin 37, and OV ± 1mV to Pins 34, 35, 41, 40, and 39; output Pins 10 and 8 shall be Logic 1. Incrementally increase Pin 39 voltage, when it exceeds +1.05V ± 100mV output Pin 10 shall switch from Logic 1 to Logic 0. Continue increasing Pin 39 voltage. Ouput Pin 8 shall switch from Logic 1 to Logic 0 when Pin 39 exceeds 4.170V ± .1V.

Return Pin 39 to OV ± 1mV. Incrementally decrease Pin 39 voltage, when Pin 39 decreases below -1.05V ± 100mV, output Pin 10 shall switch from Logic 1 to Logic 0. When Pin 39 decreases below -4.170V ± .1V, output Pin 8 shall switch from Logic 1 to Logic 0.

Return Pin 39 to OV ± 1mV. Incrementally increase Pin 35 voltage. When Pin 35 voltage exceeds .419V ± .1V, output Pin 10 shall switch from Logic 1 to Logic 0. When Pin 35 exceeds 1.67V ± .1V,output Pin 8 shall switch from Logic 1 to Logic 0.

Return Pin 35 to OV ± 1mV. Incrementally decrease Pin 35 voltage. When Pin 35 decreases below -.419V ± .1V, output Pin 10 shall switch from Logic 1 to Logic 0. When Pin 35 decreases below -1.67V ± .1V, output Pin 8 shall switch from Logic 1 to Logic 0.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 1174, 5243.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A616

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.